Rigid-body motion notes II

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1 Backward Euler with Newton's Second Law

1.1 Backward Euler Time Integration

Given position $\mathbf{x}^n$, velocity $\mathbf{v}^n$, and time step Δt, the implicit Backward Euler update is:

$$\frac{\mathbf{x}^{n+1} - \mathbf{x}^n}{\Delta t} = \mathbf{v}^{n+1}, \quad \mathbf{M}\frac{\mathbf{v}^{n+1} - \mathbf{v}^n}{\Delta t} = -\frac{\partial PE}{\partial \mathbf{x}}(\mathbf{x}^{n+1})$$

Combining the equations:

$$\mathbf{M}\left(\frac{\left(\frac{\mathbf{x}^{n+1}-\mathbf{x}^n}{\Delta t}\right) - \mathbf{v}^n}{\Delta t}\right) = -\frac{\partial PE}{\partial \mathbf{x}}(\mathbf{x}^{n+1})$$

Multiply both sides by Δt^2 and rearrange:

$$\mathbf{M}(\mathbf{x}^{n+1} - \hat{\mathbf{x}}^n) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{x}}(\mathbf{x}^{n+1}) = 0 \quad \Longleftrightarrow \quad \frac{\partial}{\partial \mathbf{x}}\Psi(\mathbf{x}) = 0$$

where

$$\hat{\mathbf{x}}^n := \mathbf{x}^n + \Delta t \mathbf{v}^n$$

and the energy-like functional is defined as:

$$\Psi(\mathbf{x}) := \frac{1}{2}(\mathbf{x} - \hat{\mathbf{x}}^n)^T \mathbf{M}(\mathbf{x} - \hat{\mathbf{x}}^n) + \Delta t^2 \cdot PE(\mathbf{x})$$

Thus, the update step is equivalent to solving the optimization problem:

$$\mathbf{x}^{n+1} = \arg\min_{\mathbf{x}} \Psi(\mathbf{x})$$

Note that in index notation:

$$\Psi(\mathbf{x}) = \frac{1}{2}(\mathbf{x}_{p\gamma} - \hat{\mathbf{x}}_{p\gamma})\mathbf{M}_{pq\gamma\epsilon}(\mathbf{x}_{q\epsilon} - \hat{\mathbf{x}}_{q\epsilon}) + \Delta t^2 \cdot PE(\mathbf{x})$$

where $\mathbf{M}$ is block diagonal, so that:

$$\mathbf{M}_{pq\gamma\epsilon} = m_p \, \delta_{pq} \, \delta_{\gamma\epsilon}$$

1.2 Gradient of the Energy Functional

Let

$$\mathbf{y}_{q\alpha}(\theta, \mathbf{c}) = \mathbf{R}_{\alpha\sigma}(\theta)\,(\mathbf{x}_{q\sigma} - \hat{\mathbf{x}}_\sigma) + \mathbf{c}_\alpha$$

where $\mathbf{c} \in \mathbb{R}^2$, $\theta \in \mathbb{R}$, and $\alpha = 1, 2$.

We now compute the gradient of the energy functional $\Psi(\mathbf{y})$ with respect to $\mathbf{y}_{p\delta}$, where:

$$\Psi(\mathbf{y}) = \frac{1}{2}(\mathbf{y}_{p\gamma} - \hat{\mathbf{x}}_{p\gamma})\mathbf{M}_{pq\gamma\epsilon}(\mathbf{y}_{q\epsilon} - \hat{\mathbf{x}}_{q\epsilon}) + \Delta t^2 \cdot PE(\mathbf{y})$$

$$\begin{aligned}\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}}(\mathbf{y}) &= \frac{\partial}{\partial \mathbf{y}_{p\delta}}\left[\frac{1}{2}(\mathbf{y}_{p\gamma} - \hat{\mathbf{x}}_{p\gamma})\mathbf{M}_{pq\gamma\epsilon}(\mathbf{y}_{q\epsilon} - \hat{\mathbf{x}}_{q\epsilon}) + \Delta t^2 \cdot PE(\mathbf{y})\right] \\ &= \mathbf{M}_{pq\delta\epsilon}(\mathbf{y}_{q\epsilon} - \hat{\mathbf{x}}_{q\epsilon}) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{y}_{p\delta}}(\mathbf{y})\end{aligned}$$

Using the fact that $\mathbf{M}$ is block diagonal, i.e.,

$$\mathbf{M}_{pq\delta\epsilon} = M_{pq}\,\delta_{\delta\epsilon},$$

we get:

$$\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}}(\mathbf{y}) = M_{pq}(\mathbf{y}_{q\delta} - \hat{\mathbf{x}}_{q\delta}) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{y}_{p\delta}}(\mathbf{y}).$$

1.3 Rigid Body Reduction and Derivatives

We define the reduced energy functional as:

$$\hat{\Psi}(\theta, \mathbf{c}) := \Psi(\mathbf{y}(\theta, \mathbf{c}))$$

Our goal is to compute gradients of $\hat{\Psi}$ with respect to the rigid-body parameters $(\theta, \mathbf{c})$.

Translational Derivative By the chain rule:

$$\frac{\partial \hat{\Psi}}{\partial \mathbf{c}_\alpha} = \frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}}(\mathbf{y}(\theta, \mathbf{c})) \cdot \frac{\partial \mathbf{y}_{p\delta}}{\partial \mathbf{c}_\alpha}$$

Let $u_p = 1$, and assume the mass matrix is diagonal: $\mathbf{M}_{pq\delta\epsilon} = m_p \delta_{pq} \delta_{\delta\epsilon}$.
From earlier, we know the gradient of Ψ is:

$$\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}} = m_p(\mathbf{y}_{p\delta} - \hat{\mathbf{x}}_{p\delta}) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{y}_{p\delta}}$$

Next, recall the rigid-body motion:

$$\mathbf{y}_{p\delta} = \mathbf{R}_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) + \mathbf{c}_\delta \quad \Rightarrow \quad \frac{\partial \mathbf{y}_{p\delta}}{\partial \mathbf{c}_\alpha} = \delta_{\delta\alpha}$$

Substituting into the chain rule:

$$\frac{\partial \hat{\Psi}}{\partial \mathbf{c}_\alpha} = \sum_p \left[m_p(\mathbf{y}_{p\alpha} - \hat{\mathbf{x}}_{p\alpha}) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{y}_{p\alpha}}\right] \cdot u_p$$

Since $u_p = 1$, we have:

$$\frac{\partial \hat{\Psi}}{\partial \mathbf{c}_\alpha} = \sum_p m_p(\underbrace{\mathbf{y}_{p\alpha}}_{\mathbf{R}_{\alpha\sigma}(\theta)(\mathbf{x}_{p\sigma}-\hat{\mathbf{x}}_\sigma)+\mathbf{c}_\alpha} - \hat{\mathbf{x}}_{p\alpha}) + \Delta t^2 \sum_p \frac{\partial PE}{\partial \mathbf{y}_{p\alpha}}$$

Note that the mass-weighted rotation term vanishes:

$$\sum_p m_p \mathbf{R}_{\alpha\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) = \mathbf{R}_{\alpha\sigma}(\theta) \sum_p m_p(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) = 0$$

So the first term simplifies to:

$$\sum_p m_p(\mathbf{c}_\alpha - \hat{\mathbf{x}}_{p\alpha}) = \left(\sum_p m_p\right) \mathbf{c}_\alpha - \sum_p m_p \hat{\mathbf{x}}_{p\alpha}$$

Let $m := \sum_p m_p$. Then the translation gradient becomes:

$$\frac{\partial \hat{\Psi}}{\partial \mathbf{c}_\alpha} = m\mathbf{c}_\alpha - \sum_p m_p \hat{\mathbf{x}}_{p\alpha} + \Delta t^2 \sum_p \frac{\partial PE}{\partial \mathbf{y}_{p\alpha}}$$

Rotation Derivative Similarly, for the rotation parameter θ, we apply the chain rule:

$$\frac{\partial \hat{\Psi}}{\partial \theta} = \frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}} \cdot \frac{\partial \mathbf{y}_{p\delta}}{\partial \theta}$$

To compute $\frac{\partial \mathbf{y}_{p\delta}}{\partial \theta}$, recall the rigid-body transformation:

$$\mathbf{y}_{p\delta} = \mathbf{R}_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) + \mathbf{c}_\delta$$

Therefore:

$$\frac{\partial \mathbf{y}_{p\delta}}{\partial \theta} = \frac{\partial \mathbf{R}_{\delta\sigma}}{\partial \theta}(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma)$$

Let $\mathbf{R}(\theta)$ be the standard 2D rotation matrix:

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad \Rightarrow \quad \mathbf{R}'(\theta) = \begin{bmatrix} -\sin\theta & -\cos\theta \\ \cos\theta & -\sin\theta \end{bmatrix}$$

Then the derivative becomes:

$$\frac{\partial \mathbf{y}_{p\delta}}{\partial \theta} = \mathbf{R}'_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma)$$

Substituting into the chain rule:

$$\frac{\partial \hat{\Psi}}{\partial \theta} = \sum_p \frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}} \cdot \mathbf{R}'_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma)$$

Using the expression for $\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}}$ from before:

$$\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}}(\mathbf{y}) = M_{pq}(\mathbf{y}_{q\delta} - \hat{\mathbf{x}}_{q\delta}) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{y}_{p\delta}}(\mathbf{y}).$$

We arrive at:

$$\frac{\partial \hat{\Psi}}{\partial \theta} = \left[M_{pq}(\mathbf{y}_{p\delta} - \hat{\mathbf{x}}_{p\delta}) + \Delta t^2 \frac{\partial PE}{\partial \mathbf{y}_{p\delta}}\right] \cdot \mathbf{R}'_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma)$$

1.4 Second Derivatives of the Reduced Energy

To apply Newton's method, we compute the Hessian of the reduced energy:

$$\nabla^2 \hat{\Psi}(\theta, \mathbf{c}) \in \mathbb{R}^{3\times 3}$$

The full Hessian consists of three blocks:

- Rotation-rotation: $\dfrac{\partial^2 \hat{\Psi}}{\partial \theta^2}$
- Rotation-translation: $\dfrac{\partial^2 \hat{\Psi}}{\partial \theta \partial \mathbf{c}_\alpha}$
- Translation-translation: $\dfrac{\partial^2 \hat{\Psi}}{\partial \mathbf{c}_\alpha \partial \mathbf{c}_\beta}$

These components form the symmetric 3×3 Hessian matrix:

$$\nabla^2 \hat{\Psi}(\theta, \mathbf{c}) = \begin{bmatrix} \dfrac{\partial^2 \hat{\Psi}}{\partial \theta^2} & \dfrac{\partial^2 \hat{\Psi}}{\partial \theta \partial \mathbf{c}_1} & \dfrac{\partial^2 \hat{\Psi}}{\partial \theta \partial \mathbf{c}_2} \\ \dfrac{\partial^2 \hat{\Psi}}{\partial \mathbf{c}_1 \partial \theta} & \dfrac{\partial^2 \hat{\Psi}}{\partial \mathbf{c}_1^2} & \dfrac{\partial^2 \hat{\Psi}}{\partial \mathbf{c}_1 \partial \mathbf{c}_2} \\ \dfrac{\partial^2 \hat{\Psi}}{\partial \mathbf{c}_2 \partial \theta} & \dfrac{\partial^2 \hat{\Psi}}{\partial \mathbf{c}_2 \partial \mathbf{c}_1} & \dfrac{\partial^2 \hat{\Psi}}{\partial \mathbf{c}_2^2} \end{bmatrix}$$

Preliminary Second Derivative Computation Taking the second derivative with respect to $\mathbf{y}_{p\delta}$ and $\mathbf{y}_{q\epsilon}$, we apply the product rule:

$$\begin{aligned} \frac{\partial^2 \Psi}{\partial \mathbf{y}_{p\delta} \partial \mathbf{y}_{q\epsilon}} &= \frac{\partial}{\partial \mathbf{y}_{q\epsilon}} \left[M_{pr}(\mathbf{y}_{r\delta} - \hat{\mathbf{x}}_{r\delta}) + \Delta t^2 \cdot \frac{\partial PE}{\partial \mathbf{y}_{p\delta}} \right] \\ &= M_{pq} \delta_{\delta\epsilon} + \Delta t^2 \cdot \frac{\partial^2 PE}{\partial \mathbf{y}_{p\delta} \partial \mathbf{y}_{q\epsilon}} \end{aligned}$$

Translation-Translation Block. We compute the second derivative of the reduced energy with respect to the translation parameters using the chain rule:

$$\begin{aligned} \frac{\partial^2 \hat{\Psi}}{\partial \mathbf{c}_\alpha \partial \mathbf{c}_\beta} &= \frac{\partial}{\partial \mathbf{c}_\beta} \left(\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}} \cdot \frac{\partial \mathbf{y}_{p\delta}}{\partial \mathbf{c}_\alpha} \right) \\ &= \frac{\partial^2 \Psi}{\partial \mathbf{y}_{p\delta} \partial \mathbf{y}_{q\epsilon}} \cdot \frac{\partial \mathbf{y}_{q\epsilon}}{\partial \mathbf{c}_\beta} \cdot \frac{\partial \mathbf{y}_{p\delta}}{\partial \mathbf{c}_\alpha} + \frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}} \cdot \frac{\partial^2 \mathbf{y}_{p\delta}}{\partial \mathbf{c}_\alpha \partial \mathbf{c}_\beta} \end{aligned}$$

Since $\mathbf{y}_{p\delta} = \mathbf{R}_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) + \mathbf{c}_\delta$, it follows that:

$$\frac{\partial \mathbf{y}_{p\delta}}{\partial \mathbf{c}_\alpha} = \delta_{\delta\alpha}, \quad \frac{\partial^2 \mathbf{y}_{p\delta}}{\partial \mathbf{c}_\alpha \partial \mathbf{c}_\beta} = 0$$

Therefore, the second term vanishes and we are left with:

$$\begin{aligned}\frac{\partial^2 \hat{\Psi}}{\partial \mathbf{c}_\alpha \partial \mathbf{c}_\beta} &= \frac{\partial^2 \Psi}{\partial \mathbf{y}_{p\delta} \partial \mathbf{y}_{q\epsilon}} \cdot \delta_{\epsilon\beta} \cdot \delta_{\delta\alpha} \\ &= \left(m_{pq}\delta_{\delta\epsilon} + \Delta t^2 \cdot \frac{\partial^2 PE}{\partial \mathbf{y}_{p\delta} \partial \mathbf{y}_{q\epsilon}} \right) \cdot \delta_{\delta\alpha} \cdot \delta_{\epsilon\beta}\end{aligned}$$

Substituting the Kronecker deltas gives:

$$\boxed{\frac{\partial^2 \hat{\Psi}}{\partial \mathbf{c}_\alpha \partial \mathbf{c}_\beta} = m_{pq}\delta_{\alpha\beta} + \Delta t^2 \cdot \frac{\partial^2 PE}{\partial \mathbf{y}_{p\alpha} \partial \mathbf{y}_{q\beta}}}$$

Rotation-Translation Block. We compute the mixed second derivative using the chain rule:

$$\begin{aligned}\frac{\partial^2 \hat{\Psi}}{\partial \theta \partial \mathbf{c}_\alpha} &= \frac{\partial}{\partial \mathbf{c}_\alpha} \left(\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}} \cdot \frac{\partial \mathbf{y}_{p\delta}}{\partial \theta} \right) \\ &= \frac{\partial^2 \Psi}{\partial \mathbf{y}_{p\delta} \partial \mathbf{y}_{q\epsilon}} \cdot \frac{\partial \mathbf{y}_{q\epsilon}}{\partial \mathbf{c}_\alpha} \cdot \frac{\partial \mathbf{y}_{p\delta}}{\partial \theta} + \frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}} \cdot \frac{\partial^2 \mathbf{y}_{p\delta}}{\partial \theta \partial \mathbf{c}_\alpha}\end{aligned}$$

Note that $\mathbf{y}_{p\delta}$ is affine in $\mathbf{c}$, and its dependence on θ and $\mathbf{c}$ is separable:

$$\mathbf{y}_{p\delta} = \mathbf{R}_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) + \mathbf{c}_\delta \quad \Rightarrow \quad \frac{\partial^2 \mathbf{y}_{p\delta}}{\partial \theta \, \partial \mathbf{c}_\alpha} = 0$$

So the second term vanishes, and we are left with:

$$\begin{aligned}\frac{\partial^2 \hat{\Psi}}{\partial \theta \partial \mathbf{c}_\alpha} &= \frac{\partial^2 \Psi}{\partial \mathbf{y}_{p\delta} \partial \mathbf{y}_{q\epsilon}} \cdot \delta_{\epsilon\alpha} \cdot \mathbf{R}'_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma) \\ &= \left(m_{pq}\delta_{\delta\epsilon} + \Delta t^2 \cdot \frac{\partial^2 PE}{\partial \mathbf{y}_{p\delta} \partial \mathbf{y}_{q\epsilon}} \right) \cdot \delta_{\epsilon\alpha} \cdot \mathbf{R}'_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma)\end{aligned}$$

Substituting the Kronecker delta simplifies this to:

$$\boxed{\frac{\partial^2 \hat{\Psi}}{\partial \theta \partial \mathbf{c}_\alpha} = \left(m_{pq}\delta_{\delta\alpha} + \Delta t^2 \cdot \frac{\partial^2 PE}{\partial \mathbf{y}_{p\delta} \partial \mathbf{y}_{q\alpha}} \right) \cdot \mathbf{R}'_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma)}$$

Rotation-Rotation Block. Using the product rule, we expand the second derivative of the reduced energy:

$$\begin{aligned}\frac{\partial^2 \hat{\Psi}}{\partial \theta^2} &= \frac{\partial}{\partial \theta} \left(\frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}} \cdot \frac{\partial \mathbf{y}_{p\delta}}{\partial \theta} \right) \\ &= \frac{\partial^2 \Psi}{\partial \mathbf{y}_{p\delta} \partial \mathbf{y}_{q\epsilon}} \cdot \frac{\partial \mathbf{y}_{q\epsilon}}{\partial \theta} \cdot \frac{\partial \mathbf{y}_{p\delta}}{\partial \theta} + \frac{\partial \Psi}{\partial \mathbf{y}_{p\delta}} \cdot \frac{\partial^2 \mathbf{y}_{p\delta}}{\partial \theta^2}\end{aligned}$$

We use the following expressions for the derivatives of the rigid-body transformation:

$$\frac{\partial \mathbf{y}_{p\delta}}{\partial \theta} = \mathbf{R}'_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma), \quad \frac{\partial^2 \mathbf{y}_{p\delta}}{\partial \theta^2} = \mathbf{R}''_{\delta\sigma}(\theta)(\mathbf{x}_{p\sigma} - \hat{\mathbf{x}}_\sigma)$$

The second derivative of the 2D rotation matrix $\mathbf{R}''(\theta) \in \mathbb{R}^{2\times 2}$ is given by:

$$\mathbf{R}''(\theta) = \begin{bmatrix} -\cos\theta & \sin\theta \\ -\sin\theta & -\cos\theta \end{bmatrix}$$

1.5 Newton's Method for Rigid-Body Optimization

We solve the nonlinear optimization problem:

$$(\theta^{n+1}, \mathbf{c}^{n+1}) = \arg\min_{\theta,\mathbf{c}} \hat{\Psi}(\theta, \mathbf{c}),$$

where $\hat{\Psi}(\theta, \mathbf{c})$ is a discrete surrogate energy under time step Δt.

Using Newton's method, we iteratively solve:

$$\nabla^2 \hat{\Psi}(\theta^k, \mathbf{c}^k) \begin{bmatrix} \Delta\theta \\ \Delta\mathbf{c} \end{bmatrix} = -\nabla \hat{\Psi}(\theta^k, \mathbf{c}^k),$$

and update:

$$\theta^{k+1} \leftarrow \theta^k + \Delta\theta, \quad \mathbf{c}^{k+1} \leftarrow \mathbf{c}^k + \Delta\mathbf{c}.$$

This continues until $\|\nabla\hat{\Psi}\|$ falls below a threshold.

Potential Energy with Circular Constraint Let $\mathbf{y}_i \in \mathbb{R}^2$ be the world position of vertex i, and let $\mathbf{c}_{\text{circle}} \in \mathbb{R}^2$ and $R > 0$ define the center and radius of a circular domain.

We define the signed distance to the boundary:

$$r_i := \|\mathbf{y}_i - \mathbf{c}_{\text{circle}}\|, \quad d_i := R - r_i,$$

and the total potential energy as:

$$\text{PE}(\mathbf{y}) = \sum_{i=1}^{4} \left[b(d_i) + mg \cdot y_{i2} \right],$$

where $b(d)$ is a barrier function that activates when a vertex nears the boundary.

We use a smooth truncated Taylor expansion of a logarithmic barrier:

$$b(d) = \begin{cases} \sum_{n=1}^{N} \frac{(-1)^n}{n\hat{d}^n}(d - \hat{d})^{n+2}, & \text{if } d < \hat{d}, \\ 0, & \text{otherwise.} \end{cases}$$

Its derivatives are:

$$b'(d) = \begin{cases} \sum_{n=1}^{N} \frac{(-1)^n (n+2)}{n\hat{d}^n}(d - \hat{d})^{n+1}, & \text{if } d < \hat{d}, \\ 0, & \text{otherwise,} \end{cases} \qquad b''(d) = \begin{cases} \sum_{n=1}^{N} \frac{(-1)^n (n+2)(n+1)}{n\hat{d}^n}(d - \hat{d})^{n}, & \text{if } d < \hat{d}, \\ 0, & \text{otherwise.} \end{cases}$$

Gradient of Potential Energy Let $\mathbf{n}_i := \frac{\mathbf{y}_i - \mathbf{c}_{\text{circle}}}{r_i}$ be the radial unit vector. The gradient w.r.t. $\mathbf{y}_i$ is:

$$\frac{\partial \text{PE}}{\partial \mathbf{y}_i} = -b'(d_i) \cdot \mathbf{n}_i + mg \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Hessian of Potential Energy The Hessian is:

$$\frac{\partial^2 \text{PE}}{\partial \mathbf{y}_i^2} = -b''(d_i) \cdot \mathbf{n}_i \mathbf{n}_i^\top + \frac{b'(d_i)}{r_i} \left(\mathbf{I} - \mathbf{n}_i \mathbf{n}_i^\top \right).$$

This structure preserves radial smoothness and regularizes tangential stiffness near the boundary.

Velocity Update (Time Step n) Given a reference configuration $\mathbf{X}_{\text{ref}} \in \mathbb{R}^{4\times 2}$, the rigid-body motion is described by:

$$\mathbf{\Phi}(\mathbf{X}, t) = \mathbf{R}(\theta(t))(\mathbf{X} - \mathbf{c}_{\text{ref}}) + \mathbf{c}(t),$$

where $\mathbf{R}(\theta) \in SO(2)$ is the rotation matrix, and $\mathbf{c}_{\text{ref}}$ is the reference center of mass.

Taking time derivative:

$$\mathbf{V}(\mathbf{X}, t) = \dot{\mathbf{\Phi}}(\mathbf{X}, t) = \dot{\theta}(t)\, \mathbf{R}'(\theta(t))(\mathbf{X} - \mathbf{c}_{\text{ref}}) + \dot{\mathbf{c}}(t).$$

Switching to Eulerian coordinates:

$$\mathbf{v}(\mathbf{x}, t) = \mathbf{V}(\mathbf{\Phi}^{-1}(\mathbf{x}, t), t) = \dot{\theta}(t)\, \mathbf{W}(\mathbf{x} - \mathbf{c}(t)) + \dot{\mathbf{c}}(t),$$

where

$$\mathbf{W} := \mathbf{R}'(\theta)\, \mathbf{R}(\theta)^\top = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

Approximating derivatives with finite differences:

$$\omega^n := \frac{\theta^{n+1} - \theta^n}{\Delta t}, \quad \dot{\mathbf{c}}^n := \frac{\mathbf{c}^{n+1} - \mathbf{c}^n}{\Delta t},$$

we get the velocity of each vertex $\mathbf{x}_i^n$ as:

$$\mathbf{v}_i^n = \omega^n \cdot \mathbf{W}(\mathbf{x}_i^n - \mathbf{c}^n) + \dot{\mathbf{c}}^n.$$